WORKED SOLUTION

STEP III 1992, Question 11

Mechanics • Collisions and Restitution

STEP III 1992, Question 11

Read the question before you go on. It is in the STEP Support Programme's question database:

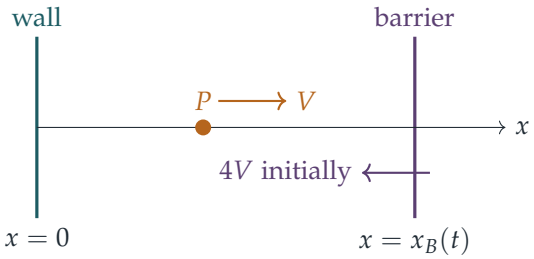
step.maths.org/questions/92-s3-q11

The key idea

Measure every position from the wall, so that the particle and the barrier each have a simple formula for their distance from it. A collision is just a moment when two of these formulas agree, which turns each timing question into a quadratic. At an impact, Newton's law of restitution applies to the *relative* velocity, so the barrier's own velocity at that instant matters.

Solution

Model P as a particle moving on a smooth table along the line through O perpendicular to the wall. Take x as distance from the wall AB, positive towards the barrier, and measure time t from the moment of projection. Velocities are positive when directed away from the wall.



The first impact

The barrier starts at distance $\frac{75}{32}VT$ with velocity $-4V$ and has constant acceleration $+4V/T$, so by the constant-acceleration formulae

$$x_B(t) = \tfrac{75}{32}VT - 4Vt + \frac{2Vt^2}{T}, \qquad v_B(t) = -4V + \frac{4Vt}{T}.$$

Until it first hits anything, P moves at constant speed (the table is smooth), so $x_P = Vt$. They meet when $x_P = x_B$. Writing $s = t/T$,

$$2s^2 - 5s + \tfrac{75}{32} = 0 \quad\Longleftrightarrow\quad 64s^2 - 160s + 75 = 0 \quad\Longleftrightarrow\quad (8s-5)(8s-15) = 0.$$

The first meeting is at the smaller root, so the impact takes place at $t = 5T/8$, as required. (Before then the gap $x_B - x_P$ is positive, since it is positive at $s = 0$ and has no root in between.)

At this moment P is at $x = \frac{5}{8}VT$ and the barrier's velocity is

$$v_B\left(\tfrac{5}{8}T\right) = -4V + \tfrac{5}{2}V = -\tfrac{3}{2}V,$$

so the barrier is still moving towards the wall, at speed $\frac{3}{2}V$.

Speed after the impact

The barrier's motion is unaffected, so only Newton's experimental law is needed. The speed of approach is $V - (-\frac{3}{2}V) = \frac{5}{2}V$. If P has velocity w just after impact, the speed of separation is $-\frac{3}{2}V - w$, and

$$-\tfrac{3}{2}V - w = \tfrac{1}{2}\cdot\tfrac{5}{2}V \quad\Longrightarrow\quad w = -\tfrac{11}{4}V.$$

So P rebounds towards the wall with speed $\dfrac{11V}{4}$.

Ask yourself Why is the rebound speed bigger than the speed V that P had before?

The barrier is moving towards P at the moment of impact, like a bat hitting a ball. Restitution acts on the speed of P *relative to the barrier*: relative to the barrier, P arrives at $\frac{5}{2}V$ and leaves at $\frac{5}{4}V$. Adding back the barrier's own $\frac{3}{2}V$ towards the wall gives $\frac{11}{4}V$.

Bounce off the wall and the second impact

P has distance $\frac{5}{8}VT$ to cover at speed $\frac{11}{4}V$, which takes $\frac{5}{8}\cdot\frac{4}{11}T = \frac{5}{22}T$. It therefore hits AB at

$$t_1 = \tfrac{5}{8}T + \tfrac{5}{22}T = \tfrac{75}{88}T.$$

The barrier cannot have got in the way. It is moving towards the wall more slowly than P throughout this interval (its speed falls from $\frac{3}{2}V$), and it never reaches the wall at all: its closest approach is at $t = T$, where $v_B = 0$ and $x_B = \frac{75}{32}VT - 4VT + 2VT = \frac{11}{32}VT > 0$.

The wall is fixed, so with $e = \frac{1}{2}$ the particle rebounds with speed $\frac{11}{8}V$, and afterwards

$$x_P = \tfrac{11}{8}V\left(t - \tfrac{75}{88}T\right) = \tfrac{11}{8}Vt - \tfrac{75}{64}VT.$$

The next collision with the barrier is at the first time after t_1 when $x_P = x_B$. In terms of $s = t/T$,

$$\tfrac{75}{32} - 4s + 2s^2 = \tfrac{11}{8}s - \tfrac{75}{64} \quad\Longleftrightarrow\quad 128s^2 - 344s + 225 = 0 \quad\Longleftrightarrow\quad (8s-9)(16s-25) = 0.$$

The roots are $s = \frac{9}{8}$ and $s = \frac{25}{16}$. Both are later than t_1, since $\frac{75}{88} < 1 < \frac{9}{8}$. At t_1 the gap $x_B - x_P = x_B(t_1)$ is positive, so the first time it closes is the smaller root. The next collision with the barrier is therefore at time $9T/8$ from the start, as required. (As a check, both are at $x = \frac{3}{8}VT$ then.)

Ask yourself Why must we take the smaller root, and not just any root?

The quadratic describes the two motions as if they could pass through each other. Its second root, $\frac{25}{16}T$, would be a "meeting" after the real collision, when the motion has already changed. The physical collision is the first time the gap closes.

Checks

- At $t = \frac{5}{8}T$: $x_B = \frac{75}{32} - \frac{80}{32} + \frac{25}{32} = \frac{20}{32} = \frac{5}{8}$ (in units of VT), equal to x_P.
- At $t = \frac{9}{8}T$: $x_B = \frac{75}{32} - \frac{144}{32} + \frac{81}{32} = \frac{12}{32} = \frac{3}{8}$, and $x_P = \frac{11}{8} \cdot \frac{24}{88} = \frac{3}{8}$.
- A time-stepping simulation with $V = T = 1$ (step 10^{-6}) found impacts at $t \approx 0.6250$ (rebound velocity -2.750), $0.8523 \approx \frac{75}{88}$ and 1.1250.

Common slips

- Applying restitution to P's speed alone, which gives $\frac{1}{2}V$ instead of $\frac{11}{4}V$. The barrier is moving, so use relative velocities.
- Getting the sign of the acceleration wrong. The barrier *starts* moving towards the wall but is decelerating, and it turns round at $t = T$.
- Forgetting to check that the barrier doesn't reach P (or the wall) before P has bounced off AB.

Beyond STEP

The neat trick here is to work in the barrier's frame at the instant of impact. Because the barrier is so massive, restitution reverses and scales the relative velocity, just as a fixed wall would. Changing frames like this is central to first-year mechanics, and it leads on to Galilean invariance and to centre-of-mass frames in two-body collisions. Kleppner and Kolenkow's *An Introduction to Mechanics* develops these ideas carefully.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.